

Adebanji Oluwatimileyin Adelowo

Data Scientist | Machine Learning Engineer | AI/LLM Engineer

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Portfolio

Education

MSc Mathematical Engineering University of L'Aquila Grade: 106/110 <i>Thesis: Mixing of Passive Scalars Advected by Incompressible Fields</i>	2018 to 2021 Italy
BSc Mathematics Obafemi Awolowo University	2014 to 2018 Nigeria

Experience

Data Scientist Intuos Srl	Dec 2024 to Sep 2025 L'Aquila
- Implemented ML classifier for flight phase identification using real-time telemetry data - Built REST APIs (50+ endpoints) for flight and pilot management systems - Built audio-based engine monitoring system using FFT for RPM detection - Developed dashboards for flight performance analysis and automated data pipelines	
Machine Learning Engineer / Data Analyst N-and Italia Srl	Apr 2023 to Nov 2024 Bergamo
- Analysed log data from 154 vending machines, processing up to 1.5M transactions per machine to extract insights on user behaviour, system performance, and failure patterns - Built dashboards tracking delivery success/failure rates, connectivity KPIs, and transaction volumes across the full machine fleet - Conducted product preference analysis across 20+ SKUs and profiled user behaviour across UI interaction stages to support UX improvements - Developed predictive models for delivery failure forecasting, identifying key pre-failure event sequences (OAA, UIPU, EIL, EID) - Investigated system reliability metrics including OS failures and mean delivery vs. selection time across 154 machines	
Operational Data Analyst eResult Srl	Jan 2023 to Mar 2023 Cesena
Optimised SQL queries and database performance; built automated dashboards and reporting systems	
Digital Engineering Intern Bridgestone EMIA	Apr 2022 to Oct 2022 Rome
- Contributed to improving Product Lifecycle Management (PLM) tools for tire development workflows - Worked on the digital twin platform used for tire engineering and simulation	
Machine Learning Intern Fellowship.AI	2022 Remote
Implemented GAN-based image generation models (InsetGAN); deployed ML models using FastAPI	

Selected Projects

Bank Customer Risk Engine: Embedding-powered risk scoring system for retail banking: encodes customer profiles into 384-dim sentence-transformer vectors, trains a multi-output model to simultaneously predict loan approval probability, default risk, and interest rate; cosine-similarity search enables nearest-neighbour profiling. *Stack: sentence-transformers, XGBoost, scikit-learn, UMAP, Python*

Enterprise NL2SQL Pipeline: Two-stage LLM pipeline converting natural language to SQL over an 8-table enterprise database, cutting prompt token cost by ~60%. Complexity router auto-escalates multi-table queries; includes retry logic and SQL injection guard. *Stack: OpenAI API (GPT-4o-mini), Python, pandas, SQLite*

Financial Sentiment Distillation: Compressed FinBERT by 62% (110M → 88M parameters) via Taylor-gradient pruning and knowledge distillation, improving accuracy from 84.5% to **96.9%** on the financial_phrasebank benchmark. *Stack: PyTorch, HuggingFace Transformers, scikit-learn*

LLM-Powered Data Analyst Agent: Autonomous data analysis agent using LangChain's Pandas DataFrame Agent; accepts natural-language instructions to clean data, identify correlations, generate visualisations, and run forecasting models (ARIMA) against a real climate dataset. *Stack: LangChain, OpenAI GPT, Pandas, Python*

Medical RAG Assistant: Conversational medical assistant over the MedQuad Q&A knowledge base; ChromaDB stores embeddings, a LangChain ReAct agent routes medical vs. general queries, and ConversationBufferWindowMemory maintains context across turns. *Stack: LangChain, ChromaDB, OpenAI, HuggingFace*

LLM Evaluation with LangSmith: Systematic evaluation framework comparing T5-base, fine-tuned T5, and OpenAI GPT on CNN/DailyMail summarisation using LangSmith tracing and cosine embedding distance; shows domain fine-tuning closes most of the gap with proprietary models. *Stack: LangSmith, LangChain, HuggingFace T5, OpenAI*

LoRA and QLoRA Fine-Tuning: Parameter-efficient fine-tuning of Llama-3.2-1B (LoRA) and Llama-3-8B (QLoRA with 4-bit NF4 quantisation); trains less than 1% of model weights, enabling 8B-parameter fine-tuning on a single 16 GB GPU. *Stack: PEFT, BitsAndBytes, HuggingFace Transformers, trl*

Skills

Data Analysis & BI

Power BI, SQL, pandas, NumPy
matplotlib, seaborn, statistical analysis
Dashboard design, KPI reporting

Machine Learning

PyTorch, TensorFlow, scikit-learn
Predictive modelling, classification, time series
Feature engineering, model evaluation

Generative AI & LLMs

LangChain, LangSmith, HuggingFace Transformers
OpenAI, AWS Bedrock, Google Vertex AI
RAG, Prompt Engineering, Fine-tuning, PEFT

Software Engineering

Python, SQL, FastAPI, Flask
REST APIs, Microservices, Git, Docker

Cloud & MLOps

AWS, Vertex AI, Model deployment, LLMOps

Vector Databases

Pinecone, FAISS, ChromaDB